

Metric Justification & Clinical Relevance

1. Dataset Recap

The models below were trained on the cleaned emergency department triage dataset which was produced in week 5 which had 55,121 encounters. The target Emergency Severity Index (ESI) is a 5 level scale which is assigned by a triage nurse. ESI 1 is the most critical (resuscitate) and

ESI 5 is the least (non-urgent). The dataset is severely imbalanced: ESI 1 makes up roughly 0.1% of encounters, with the bulk of patients falling into ESI 2–4. Data was split 80/20, stratified on ESI, with a fixed random seed (random_state=42) so the split is reproducible.

2. Model Descriptions

Two baseline models were trained, alongside a stratified random “dummy” classifier used as the floor that both baseline models must beat.

Logistic Regression

This model functions as a classifier. Each patient can be represented as a point positioned according to their vitals and other measured features, and the model estimates a boundary that separates patients belonging to different ESI levels. Each feature is assigned a weight reflecting how strongly it pushes a prediction toward a particular class, and these weighted features are combined to calculate a probability for each ESI level. This approach is efficient and its weights are directly interpretable, but it can only represent relationships that are approximately linear. Because the model is sensitive to the scale of its inputs, all features were standardized before training so that a feature such as heart rate would not dominate a feature such as temperature simply due to its larger numerical range.

Decision Tree

This model functions like a flowchart composed of a sequence of yes or no questions, for example asking whether oxygen saturation is below 93 percent before checking respiratory rate. At each step, the model selects the question that best separates patients by urgency, continuing until it reaches a final classification. It does not require feature scaling and produces a structure that can be traced step by step to explain its reasoning. Because a tree allowed to grow too deep tends to memorize the training data rather than learn generalizable rules, its maximum depth was limited to five levels,

trading a small amount of potential accuracy for a model that generalizes better and remains easy to explain.

Dummy Classifier (baseline)

This model does not learn from patient characteristics and instead serves as a point of comparison. It calculates how frequently each ESI level appears in the training data and generates predictions in those same proportions regardless of the patient being evaluated. If the trained models cannot outperform this baseline, they have not learned any meaningful pattern from the data, corresponding to the "coin flip" comparison referenced in the assignment brief.

3. Benchmark Table

Model	Accuracy	Macro F1	Recall (ESI 1)
Dummy (stratified baseline)	0.375	0.204	0.00
Logistic Regression	0.667	0.495	0.25
Decision Tree	0.556	0.216	0.00

Logistic Regression weighted F1 is 0.661 vs macro F1: 0.495. The gap between these two matters is that the weighted F1 is pulled up by the model's strong performance on the common ESI 2 and 3 classes, while the macro F1 which treats every ESI level equally exposes how much worse the model does on the rare classes, including ESI 1. Reporting weighted F1 alone would overstate how well the model serves the patients who need it the most.

4. Metric Justification

Accuracy is not a defensible primary metric for this problem. ESI 1 patients make up roughly 0.1% of the dataset, a model can score over 90 % accuracy even if the correct critical patient is identified which is exactly what the Decision Tree does here. The ESI 1 is used as the primary metric instead, because it directly measures the one outcome that matters clinically which are of the patients who needed immediate resuscitation. A missed ESI 1 patient (a false negative) means a resuscitate level case is triaged as lower priority, risking delayed care. A false alarm (flagging a stable patient as ESI 1) is

costly but can be recovered easily, the clinical staff can downgrade on reassessment. The asymmetry between these two error types is why recall, not accuracy or even precision, is the number this report leads with.

5. Failure Modes

Logistic Regression catches 4 of 16 truly critical patients (25% recall). Most misses are not scattered randomly, the confusion matrix shows that missed ESI 1 patients are entirely reclassified as ESI 2, not further away. This however, limits, but does not eliminate the patient. The clinical risk is a patient that is triaged as “emergent” instead of “resuscitate” still receives urgent attention, just not the fastest possible response. Decision Tree catches 0 of 16 critical patients with a 0% recall. At a max depth of 5, the tree does not learn any split that isolates the ESI 1 class, with only a small amount of ESI 1 examples in the training data, the tree’s splits are dominated by the larger ESI 2 populations, and ESI 1 patients simply fall through to whichever leaf their other features happen to match. This case of class imbalance makes the model less efficient which otherwise looks reasonable on accuracy (0.556) and beats the dummy baseline overall.

Which patient does this model miss? Based on the confusion matrix, the model’s ESI 1 misses are patients whose vitals and chief complaints resemble the much more common ESI 2 population. Therefore, critical patients who do not present with the most extreme vital sign abnormalities. This matters clinically because it suggests the model is not failing randomly, but specifically on the borderline critical presentations, which are the most important cases to catch.

6. Next Steps

- Address class imbalance directly. Both baselines were trained on data where ESI 1 makes up roughly 0.1% of the encounters, the model can ignore that class entirely and barely affect its overall score which is the case in this week's baseline. There concrete fixes can be tested: class weighting (class_weight = “balance” in scikit-learn), which would penalize an ESI-1 mistake more heavily than a common class mistake during training. Oversampling with SMOTE, which would generate synthetic ESI 1 training examples so the model sees more of the rare class rather than a handful of real cases, and threshold tuning, which lowers the confidence which requires before the model flags a patient as ESI 1 which

trades some additional false alarms for a higher chance of catching a true critical case.

- Evaluate more expressive models. Logistic Regression and a depth 5 Decision Tree were chosen deliberately as simple, interpretable baselines. This report documents exactly where they fail, which misses 3 of every 4 critical patients. The next step would be to test models capable of learning more complex, non-linear patterns in the data.
- Re-run the same benchmark table after every change with each future model. Whether it uses class weighting, oversampling, or a different algorithm, it will be scored on the same columns used in this report. This would keep every comparison honest and prevent a higher accuracy score from masking a model that still misses the critical patients.